# DILLAN IMANS

## Computer Vision Researcher

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### PROFILE

M.Sc. student in AI in Biomedicine at TUM and computer vision researcher with over two years of experience in medical imaging and surgical scene understanding. Contributed to four publications on domain adaptation, cross-modal learning, and multimodal perception. Interested in self-supervised multimodal representations and world models.

### EDUCATION

**M.Sc. AI in Biomedicine, Technical University of Munich** **Oct 2026 – present**
Member of Elite Network of Bavaria, Imaging and Sensing focus

**B.S. Computer Science & Engineering, Sungkyunkwan University** **Aug 2022 - Jun 2026**
95.7 / 100 · Dean's List · Academic Excellence Scholarship

### PUBLICATIONS

*Full list: scholar.google.com/citations?user=o-TtUq0AAAAJ*

**First author**

1. **Imans, D.**, Bui, P.N., Le, D.T., & Choo, H. "Clinical Graph-Mediated Distillation for Unpaired MRI-to-CFI Hypertension Prediction." *EMA4MICCAI 2026*.
2. **Imans, D.**, Bui, P.N., Le, D.T., & Choo, H. "Unsupervised Domain Adaptation with SAM-RefiSeR for Enhanced Brain Tumor Segmentation." *IEEE BIBM 2025*.
3. ***Imans, D.**, Abuhmed, T., Alharbi, M., & El-Sappagh, S. "Explainable Multi-Layer Dynamic Ensemble Framework Optimized for Depression Detection and Severity Assessment." Diagnostics, 14(21), 2385 (2024)*.
4. **Imans, D.**, Bui, P.N., Le, D.T., & Choo, H. "Shared Semantic Codebook Distillation for Unpaired Cross-Modal Medical Classification." *Preprint*.

**Co-authored**

5. Xu, M., Shen, D., Zhang, J., Yip, H.C., Gao, Y., Chen, C., **Imans, D.**, et al. "Generalized Recognition of Basic Surgical Actions Enables Skill Assessment and Vision-Language-Model-based Surgical Planning." *Accepted, npj Digital Surgery*.
6. He, Y., Guo, P., Xu, M., Li, Z., **Imans, D.**, et al. "Cosmos-H-Surgical: Learning Surgical Robot Policies from Videos via World Modeling." *Preprint*.
7. Xu, M., Huang, Z., **Imans, D.**, Ye, Y., Zhang, X., & Dou, Q. "SAP-Bench: Benchmarking Multimodal Large Language Models in Surgical Action Planning." *Preprint*.

### EXPERIENCE

**Graduate Computer Vision Researcher** **Oct 2026 - Present**
CAMP, TUM, Germany

- Developing a self-supervised multimodal OR encoder with a graph JEPA architecture and higher-order graph representations, fusing multi-view video, 3D point clouds, and audio for surgical scene understanding and scene graph generation

**Undergraduate Computer Vision Researcher** **Oct 2024 - Jun 2026**
Superintelligence Lab, SKKU, South Korea

- Developed a two-phase unsupervised domain adaptation framework (frequency alignment + SAM-guided refinement) for cross-modality brain tumor segmentation; +9.0 Dice over strongest baseline across 6 BraTS 2021 settings [2]
- Built a graph-mediated distillation framework for unpaired MRI-to-fundus hypertension prediction across disjoint cohorts; +5.2 AUC, +3.2 AUPRC over best baseline, MRI-free at inference [1]
- Built a codebook-based distillation framework for unpaired cross-modal classification; +5.7 and +2.5 macro-F1 on OCT→fundus and CT→X-ray [4]
- Built an end-to-end retinal screening system (optic disc/cup segmentation, vessel analysis) and its backend; ONNX-optimized for low-latency inference and delivered to a clinical partner

**Undergraduate Computer Vision Researcher** **Jun 2024 - Jun 2026**
Labren, CUHK, Hong Kong

- Curated a dataset and ran initial model experiments for a multimodal surgical robotics paper on learning robot policies via world modeling, in collaboration with NVIDIA [6]
- Built preprocessing infrastructure and an annotation QA pipeline for the first large-scale surgical action recognition dataset (11,915 clips) [5]
- Curated a dataset for multimodal LLM benchmarking for surgical action planning [7]

**Undergraduate Machine Learning Researcher** **Feb 2024 - Oct 2024**
Infolab, SKKU, South Korea

- Built an explainable multi-layer dynamic ensemble for depression detection and severity assessment; 88.3% accuracy across 2,763 clinical records with SHAP interpretability, benchmarked against 20+ algorithms [3]

## PROJECTS

**Haptic Needle Teleoperation** **Oct 2026 - Present**
Hexlab, TUM, Germany

- Teleoperation setup in which a Force Dimension haptic device drives a Franka robot arm carrying a needle, aimed at needle-insertion tasks (early stage)
- Adding multimodal operator feedback (haptic, vibration, audio, tongue-based tactile) so the operator can perceive what the needle is doing; working in ROS and C++

## TECHNICAL SKILLS

**Programming Languages:** Python · C/C++ · Java · TypeScript
**ML & Vision:** PyTorch · OpenCV · HuggingFace · MONAI · nnU-Net · SimpleITK
**Tooling:** Git · Docker · Linux · GitHub Actions · wandb

## SERVICE & LANGUAGES

Reviewer, EMA4MICCAI 2026 (MICCAI workshop)
English, Indonesian (Fluent) · Mandarin, Korean (Conversational) · German (B1)